Patrick Zager
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EDUCATION

University of California San Francisco **Sept. 2020 – Present**
Pursuing: Doctor of Philosophy in Biophysics *San Francisco, CA*

Lehigh University **Aug. 2013 – Aug. 2017**
Bachelor of Science with Honors *Bethlehem, PA*
Major: Bioengineering: Biomechanics and Biomaterials

RESEARCH EXPERIENCE

University of California San Francisco **Sept. 2020 – Present**
Graduate Student *San Francisco, CA*
Advisor: Orion Weiner, Ph.D, Professor Cardiovascular Research Institute

In my thesis work I am studying neutrophil membrane organization and mechanics to understand how neutrophil shape enables their rapid polarization and migration. I utilize lightsheet microscopy, optogenetic control of cell polarization, and geometric analysis of cell morphology to precisely quantify membrane dynamics. I've determined that the neutrophil membrane reservoir maintains a homeostatic membrane tension, enables faster shape change commensurate to cell reservoir size, and facilitates a rapid noncommittal mode of cell migration.

Weill Cornell Medicine **June 2018 - Sept. 2020**
Research Technician II *New York, NY*
Advisor: Enrique Rodriguez-Boulan, M.D., Professor of Cell Biology in Ophthalmology

Work researching epithelial polarity. I primarily studied the nurse-patient relationship between photoreceptors and the retinal pigment epithelium (RPE). Other work included: analysis of the polarized MDCK surface proteome, RNA-seq of RPE under degenerative conditions, and the development of CRISPR edited MDCK cell lines.

Hospital for Special Surgery **Sept. 2017 - May 2018**
Research Technician I *New York, NY*

Work researching the cellular and molecular mechanisms/mechanics of skeletal muscle and tendon development and rehabilitation. I built a single muscle fiber pulling system for testing molecular effects on muscle mechanics.

Lehigh University **May. 2017 - Aug. 2017**
Undergraduate Researcher *Bethlehem, PA*
Advisor: Hannah Dailey, Ph.D, Assistant Professor Mechanical Engineering and Mechanics

A short computational research project on the fracture healing mechanics of the tibia under Dr. Hannah Dailey. I created finite element models of fracture healing implemented through fuzzy logic of the underlying biological processes. Work awarded 1st place in the 2018 ORS International Section of Fracture Repair poster category.

PUBLICATIONS

De Belly, H., Gallén, A. F., Strickland, E., Estrada, D. C., Godinez, D. S., Neiva, E., ... & Weiner, O. D. (2026). Long-range mutual activation establishes Rho and Rac polarity during cell migration. *Nature cell biology*, *28*(6), 1244-1257.

Strickland, E. D., Kandaswamy, R., **Zager, P. J.**, de Belly, H., Phillips, T. K., Hopke, A., & Weiner, O. D. (2026). Arachidonic acid availability controls neutrophil swarm initiation and scaling. *bioRxiv*.

De Belly, H., Yan, S., da Rocha, H. B., Ichbiah, S., Town, J. P., **Zager, P. J.**, ... & Weiner, O. D. (2023). Cell protrusions and contractions generate long-range membrane tension propagation. *Cell*, *186*(14), 3049-3061.

Lehmann, G. L., Hanke-Gogokhia, C., Hu, Y., Bareja, R., Salfati, Z., Ginsberg, M., ... & Benedicto, I. (2020). Single-cell profiling reveals an endothelium-mediated immunomodulatory pathway in the eye choroid. *Journal of Experimental Medicine*, *217*(6).

Jonker, C. T., Deo, C., **Zager, P. J.**, Tkachuk, A. N., Weinstein, A. M., Rodriguez-Boulan, E., ... & Schreiner, R. (2020). Accurate measurement of fast endocytic recycling kinetics in real time. *Journal of cell science*, *133*(2).

Giovinazzo, J. A., Thomson, R. P., Khalizova, N., **Zager, P. J.**, Malani, N., Rodriguez-Boulan, E., ... & Schreiner, R. (2020). Apolipoprotein L-1 renal risk variants form active channels at the plasma membrane driving cytotoxicity. *Elife*, *9*.

Gravotta, D., Perez Bay, A., Jonker, C. T., **Zager, P. J.**, Benedicto, I., Schreiner, R., ... & Rodriguez-Boulan, E. (2019). Clathrin and clathrin adaptor AP-1 control apical trafficking of megalin in the biosynthetic and recycling routes. *Molecular biology of the cell*, *30*(14), 1716-1728.

Caceres, P. S., Gravotta, D., **Zager, P. J.**, Dephoure, N., & Rodriguez-Boulan, E. (2019). Quantitative proteomics of MDCK cells identify unrecognized roles of clathrin adaptor AP-1 in polarized distribution of surface proteins. *Proceedings of the National Academy of Sciences*, *116*(24), 11796-11805.

FELLOWSHIPS, AWARDS, AND TALKS

FELLOWSHIPS

American Heart Association Predoctoral Fellowship 2024 - 2026
NIH Research Supplement to Promote Diversity in Health-Related Research 2024 - 2026
NIH/NIBIB T32 Predoctoral Fellowship, Integrative Program in Complex Biological Systems 2023 - 2024

AWARDS

UCSF Mel Jones Award (Best 5th Year Biophysics Graduate Student Presentation) 2024

TALKS

Cold Spring Harbor Laboratory, Cell Modeling in Space and Time 2026
Gordon Research Seminary. Directed Cell Migration 2025

TEACHING EXPERIENCE

Marine Biological Laboratory **June 2026**
Physiology Course Research Facilitator (TA) *Woods Hole, MA*

Worked as a teaching assistant under Tanner Fadero for the microscopy skills week for the 2026 Physiology course. l lead small groups in building a breadboard TIRF microscope from scratch. Each group assembled and imaged microtubules on microscopes assembled over two days.

University of California San Francisco **Aug. 2021**
Biophysics Bootcamp Teaching Assistant *San Francisco, CA*

I developed and led an introduction to scientific programming course for incoming graduate PhD students. Students were introduced to programming fundamentals, editors and IDEs, version control, data analysis, and example use cases of programming in the life sciences.

Weill Cornell Medicine **Apr. 2018 - June 2018**
BioVenture eLab Instructor *New York, NY*

I developed a short course to introduce medical school students and university professors to computer-aided design. Primary outcomes included developing students' ability to rapidly prototype concepts, reverse engineer products, and design biomedical products. Final projects have students develop tools to be utilized for their research.

Lehigh University **Aug. 2015 – Dec. 2015, Aug. 2016 - Dec. 2016**
Teaching Assistant *Bethlehem, PA*

I was a teaching assistant for Introduction to Engineering and Introduction to Computer Aided-Design courses. I assisted students during and outside of class, led classes, created projects, and created exams.